NAME : Shri Harshini Devi . A

REG.NO : 192524104

Dept. Name : B.Tech-AIDS

Course Code/Course Name: DSA0602 /Data Handling and Visualization

Date: 29/07/2026

LIST OF EXPERIMENTS

Set 18: Product Inventory Management

Dataset: Product Inventory

Product ID	Product Name	Quantity Available	Price (in $)
1	Product A	250	20
2	Product B	175	15
3	Product C	300	18

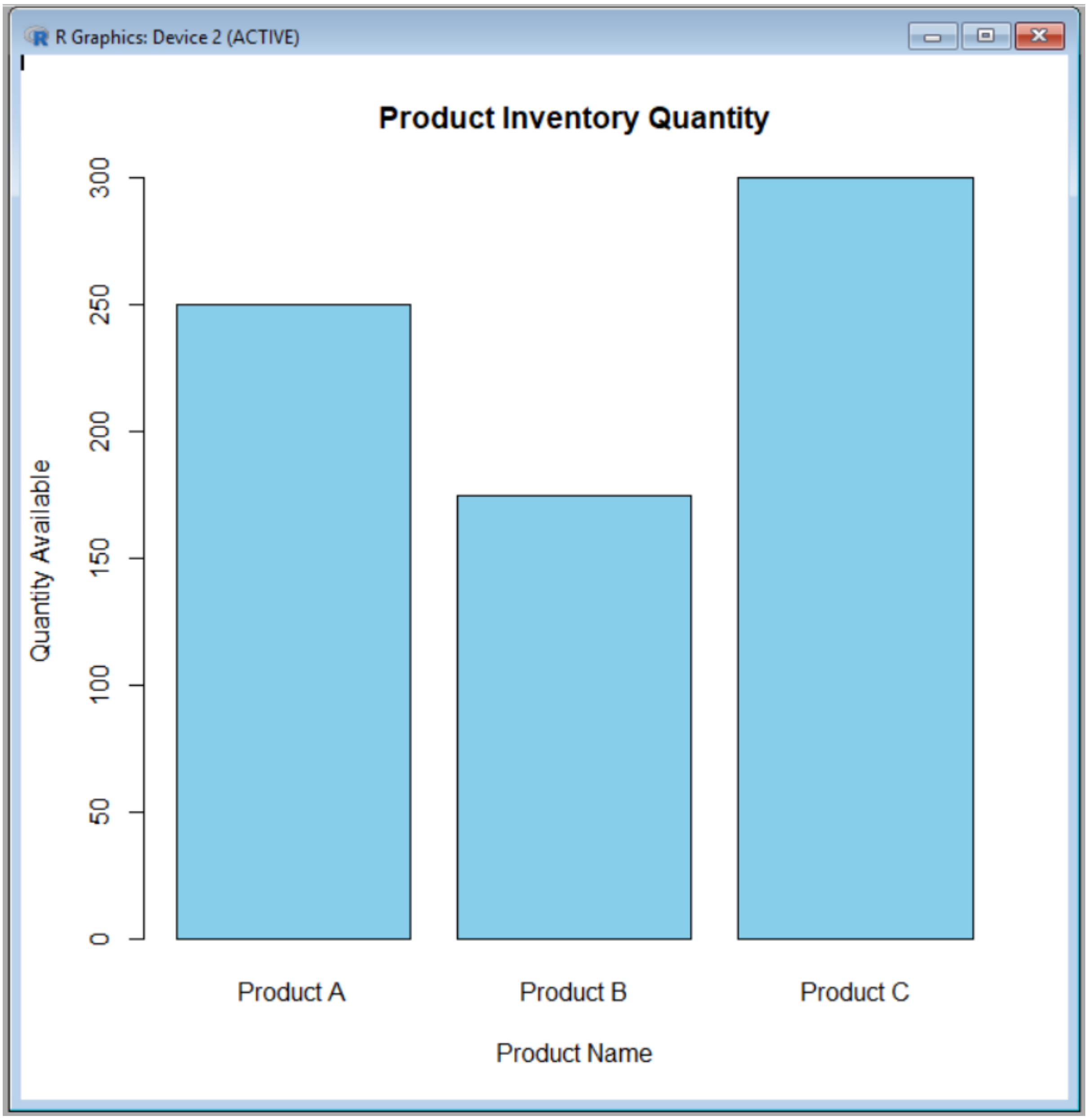
1. **Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.**

```
# Product Inventory Data

product <- c("Product A", "Product B", "Product C")

quantity <- c(250, 175, 300)


# Create Bar Chart

barplot(quantity,

        names.arg = product,

        col = "skyblue",
```

xlab = "Product Name",

ylab = "Quantity Available",

main = "Product Inventory Quantity")



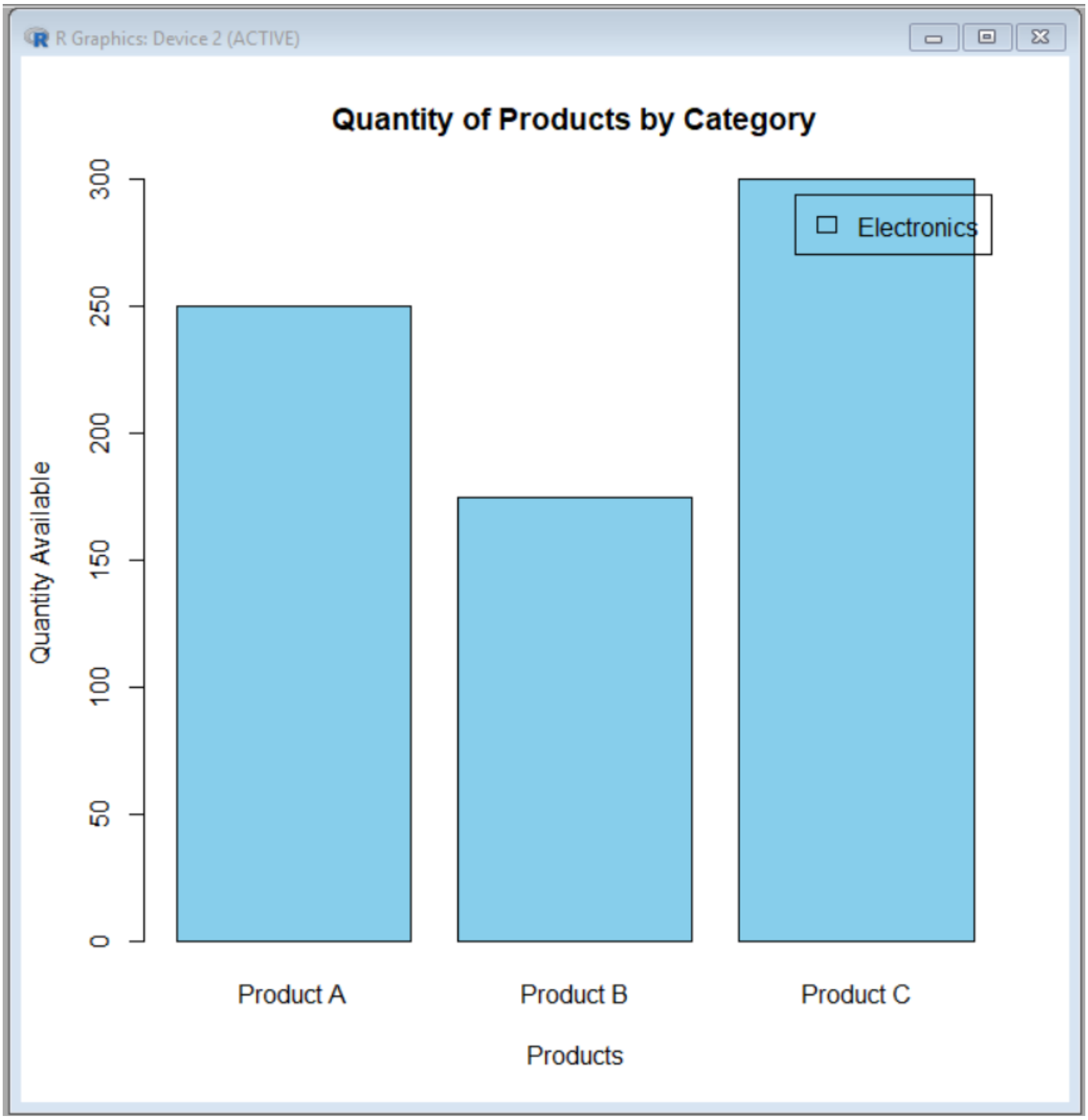
2. **Generate a stacked bar chart to show the quantity of each product within different product categories.**

Product Inventory Data

quantity <- matrix(c(250, 175, 300), nrow = 1)

Category Name

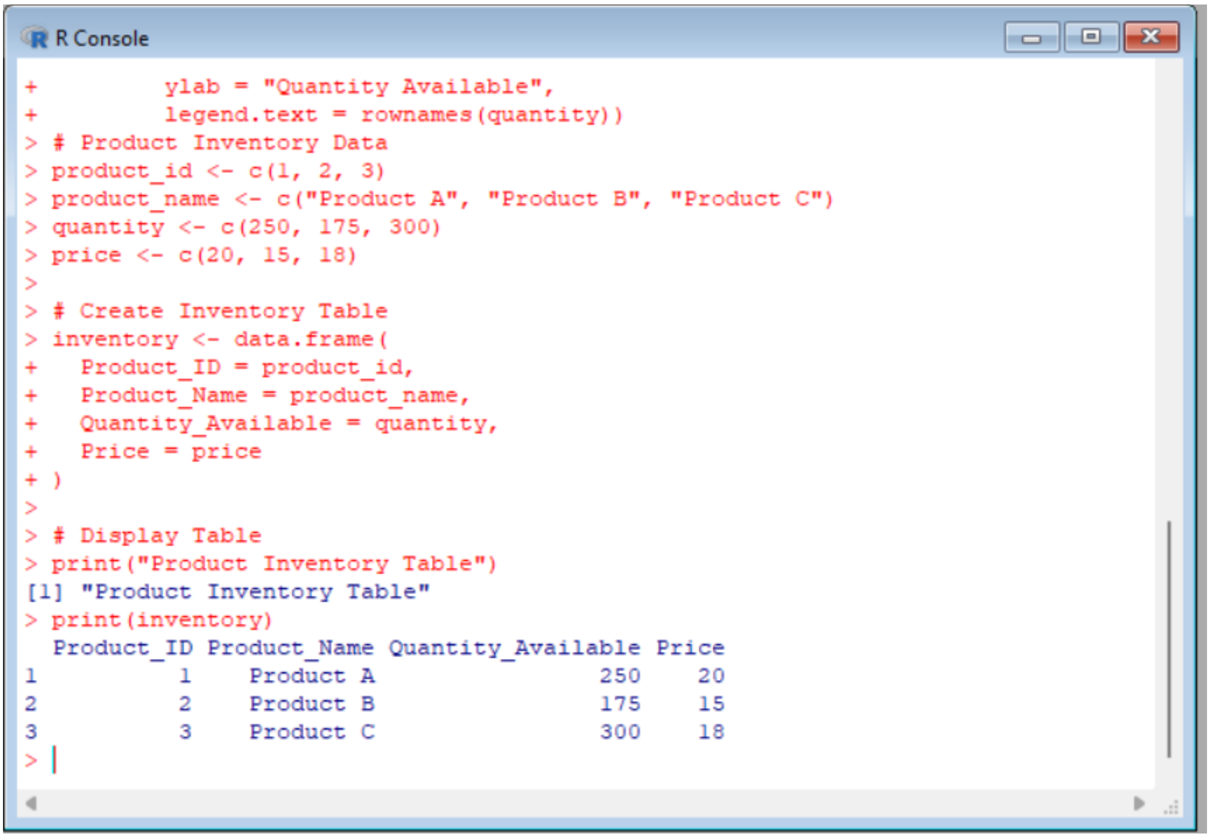
```r
rownames(quantity) <- c("Electronics")


# Product Names

colnames(quantity) <- c("Product A", "Product B", "Product C")


# Create Stacked Bar Chart

barplot(quantity,

        col = c("skyblue"),

        main = "Quantity of Products by Category",

        xlab = "Products",

        ylab = "Quantity Available",

        legend.text = rownames(quantity))
```



3. **Build a table to show the inventory data for each product. Label the table elements.**

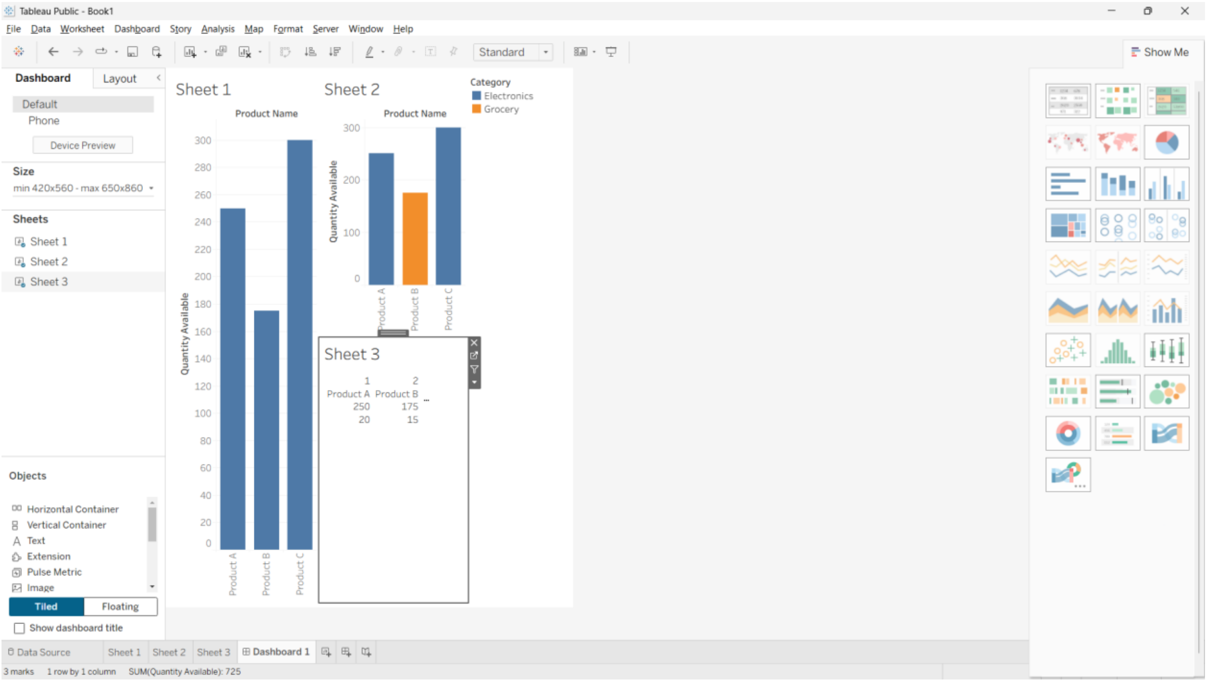
```
# Product Inventory Data
product_id <- c(1, 2, 3)
product_name <- c("Product A", "Product B", "Product C")
quantity <- c(250, 175, 300)
price <- c(20, 15, 18)

# Create Inventory Table
```

```
inventory <- data.frame(
  Product_ID = product_id,
  Product_Name = product_name,
  Quantity_Available = quantity,
  Price = price
)

# Display Table
print("Product Inventory Table")
print(inventory)
```



```
R Console
+         ylab = "Quantity Available",
+         legend.text = rownames(quantity))
> # Product Inventory Data
> product_id <- c(1, 2, 3)
> product_name <- c("Product A", "Product B", "Product C")
> quantity <- c(250, 175, 300)
> price <- c(20, 15, 18)
>
> # Create Inventory Table
> inventory <- data.frame(
+   Product_ID = product_id,
+   Product_Name = product_name,
+   Quantity_Available = quantity,
+   Price = price
+ )
>
> # Display Table
> print("Product Inventory Table")
[1] "Product Inventory Table"
> print(inventory)
  Product_ID Product_Name Quantity_Available Price
1          1    Product A                250    20
2          2    Product B                175    15
3          3    Product C                300    18
>
```

4. **Develop a Tableau dashboard combining the bar chart, stacked bar chart, and the table for interactive exploration of inventory data.**



Set 19: Survey Responses Analysis

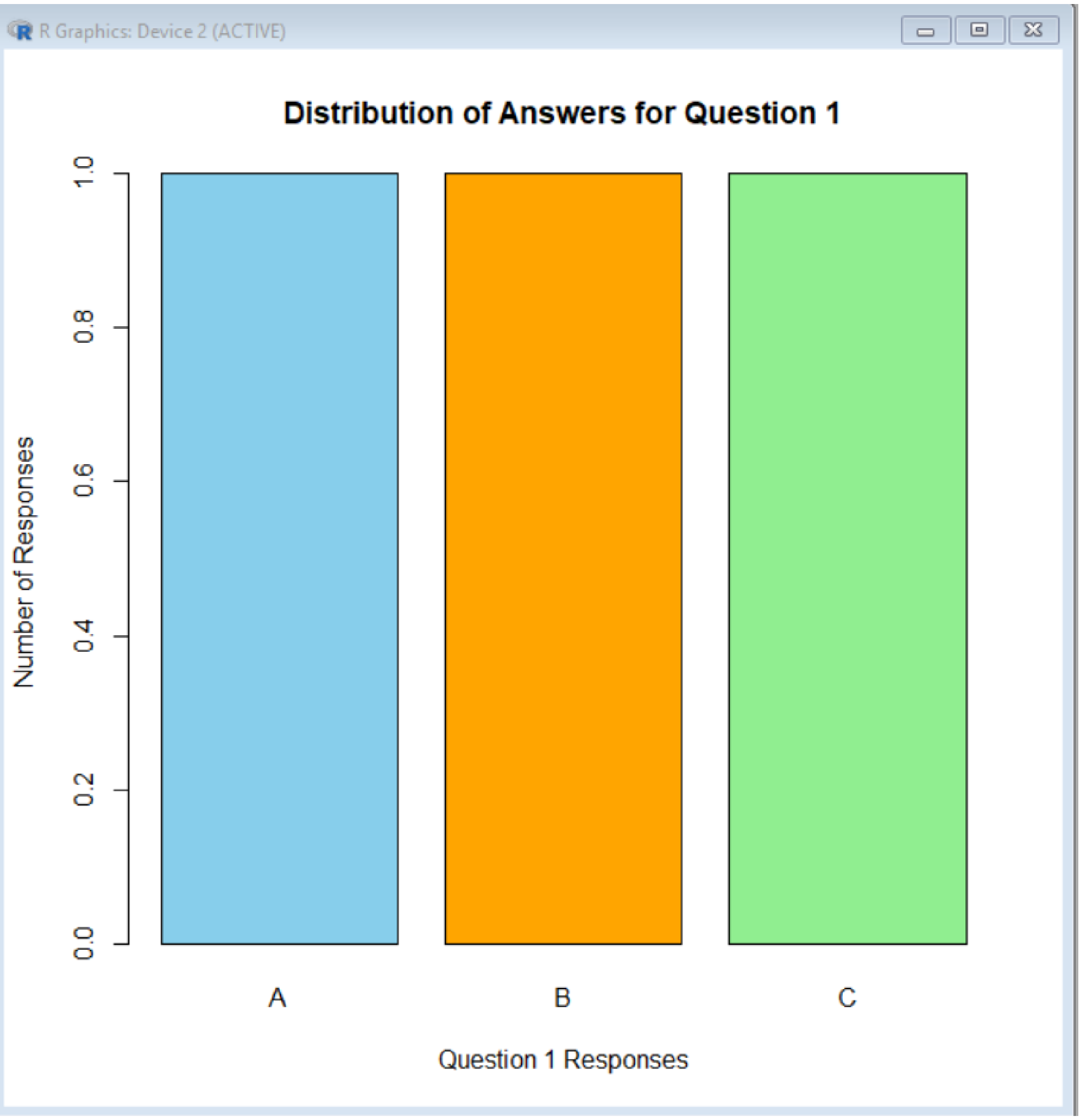
Dataset: Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

1. **Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.**

```
# Survey Responses Data

q1 <- c("A", "B", "C")


# Count the responses

count <- table(q1)


# Create Grouped Bar Chart

barplot(count,
```

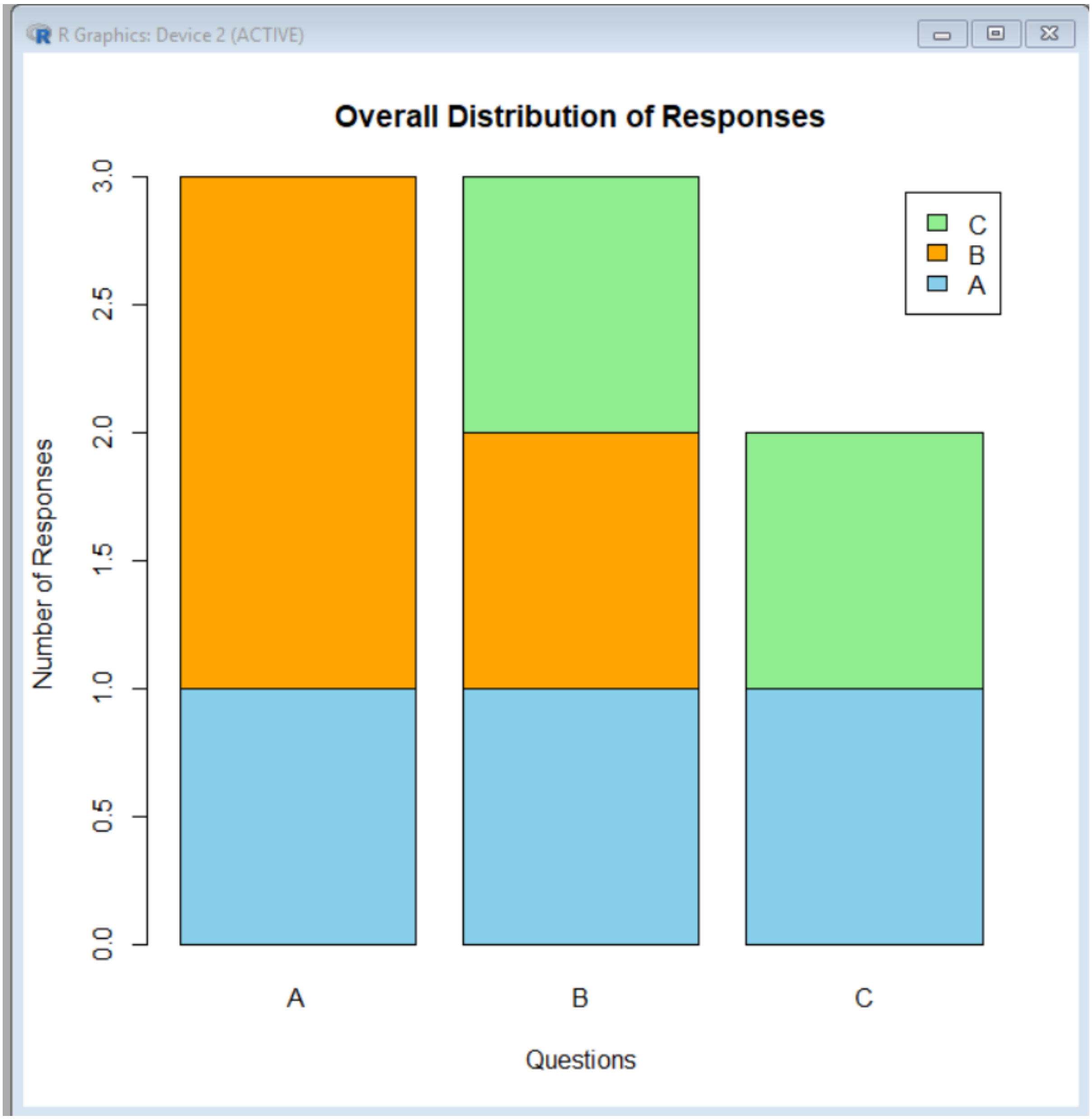
```
col = c("skyblue", "orange", "lightgreen"),
main = "Distribution of Answers for Question 1",
xlab = "Question 1 Responses",
ylab = "Number of Responses",
beside = TRUE)
```



2. **Generate a stacked bar chart to represent the overall distribution of responses for all three questions.**

```
# Survey Responses Data
responses <- matrix(c(
```

```
  1, 1, 1,   # Question 1 (A, B, C)
  2, 1, 0,   # Question 2 (A, B, C)
  0, 1, 1    # Question 3 (B, C, D)
), nrow = 3, byrow = TRUE)

# Row and Column Names
rownames(responses) <- c("Question 1", "Question 2", "Question 3")
colnames(responses) <- c("A", "B", "C")

# Create Stacked Bar Chart
barplot(responses,
        col = c("skyblue", "orange", "lightgreen"),
        main = "Overall Distribution of Responses",
        xlab = "Questions",
        ylab = "Number of Responses",
        legend.text = colnames(responses))
```



3. **Build a table to show the survey response data for each survey. Label the table elements.**

Survey Response Data

survey_id <- c(1, 2, 3)

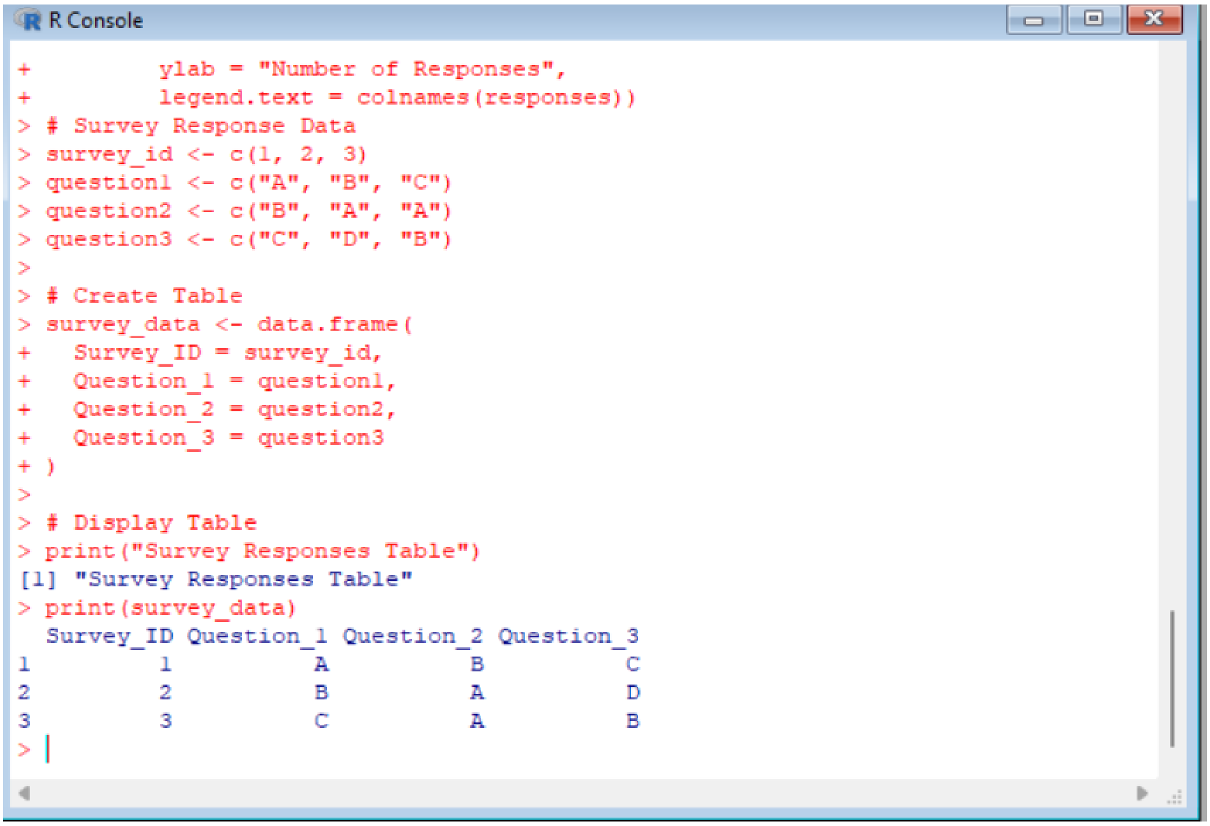
question1 <- c("A", "B", "C")

question2 <- c("B", "A", "A")

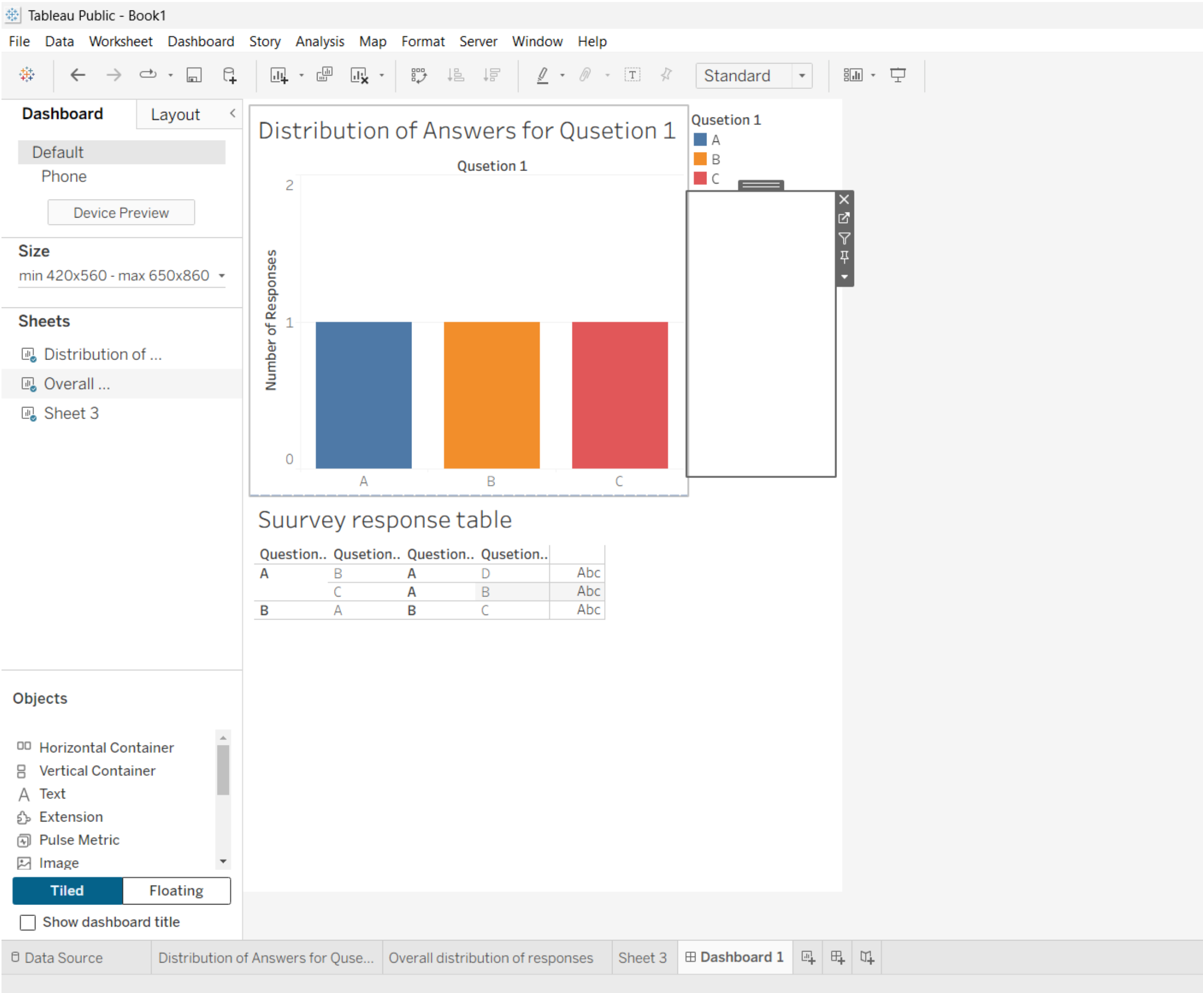
question3 <- c("C", "D", "B")

Create Table

```
survey_data <- data.frame(
  Survey_ID = survey_id,
  Question_1 = question1,
  Question_2 = question2,
  Question_3 = question3
)

# Display Table
print("Survey Responses Table")
print(survey_data)
```



4. **Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.**



Set 20: Stock Analysis

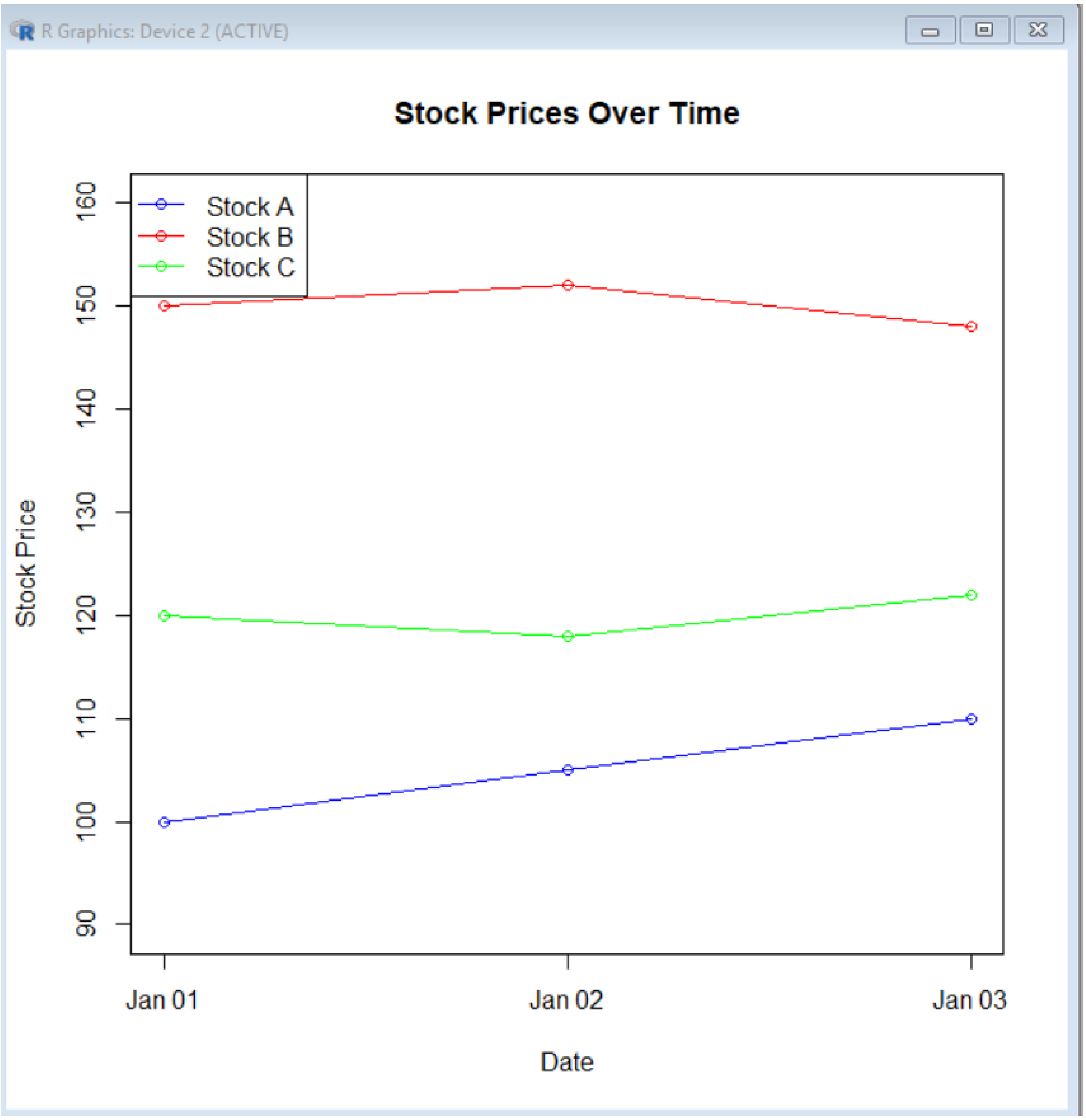
Dataset: Stock Prices

2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122

1. **Create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock C) over a specific time period. Label the axes and title the chart.**

```
# Stock Analysis Dataset


date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03"))

stockA <- c(100, 105, 110)
```

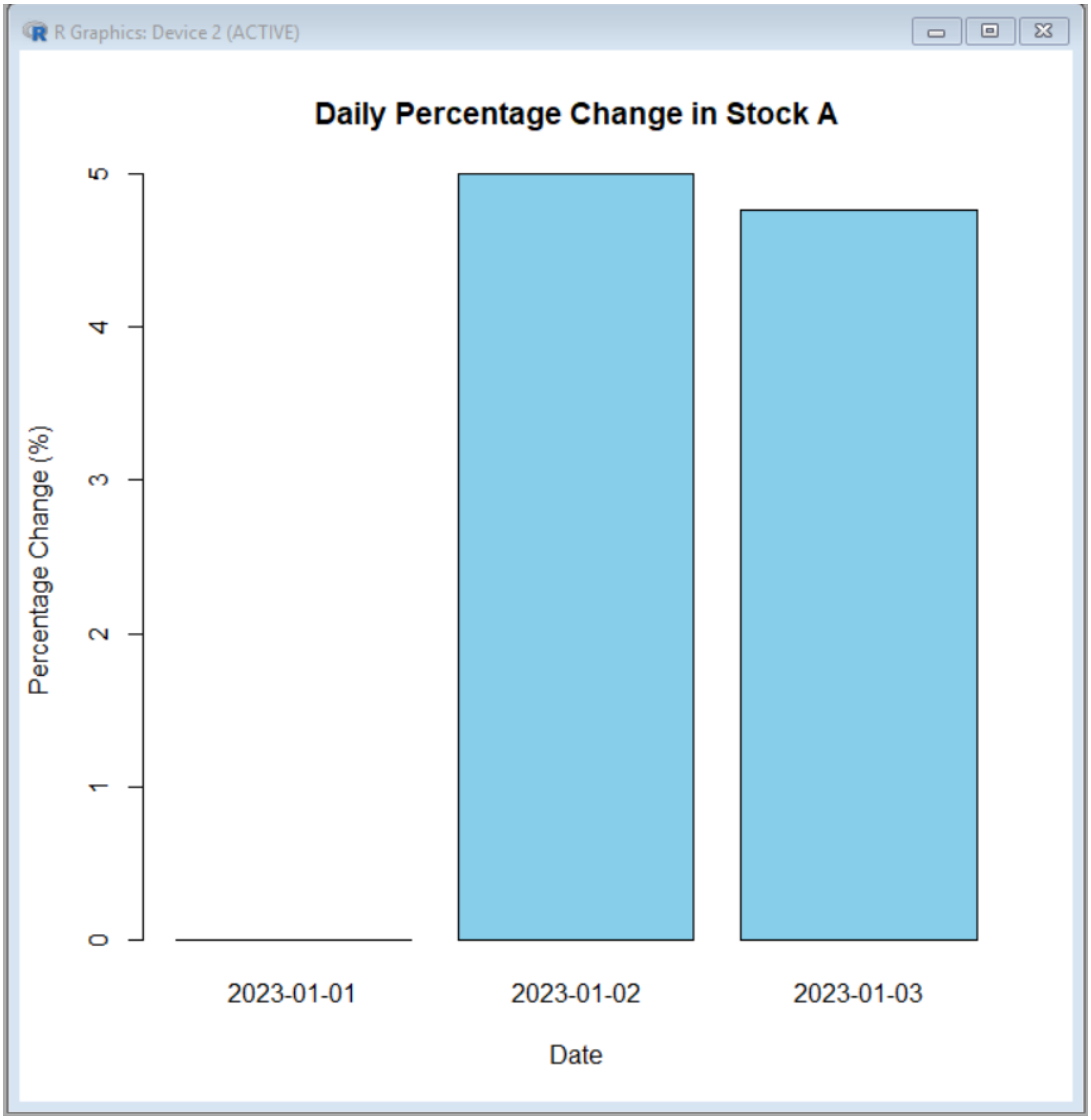
```
stockB <- c(150, 152, 148)
stockC <- c(120, 118, 122)

# Plot Stock A
plot(date, stockA, type = "o", col = "blue",
    ylim = c(90,160),
    xlab = "Date",
    ylab = "Stock Price",
    main = "Stock Prices Over Time")

# Add Stock B
lines(date, stockB, type = "o", col = "red")

# Add Stock C
lines(date, stockC, type = "o", col = "green")

# Add Legend
legend("topleft",
      legend = c("Stock A", "Stock B", "Stock C"),
      col = c("blue", "red", "green"),
      lty = 1,
      pch = 1)
```



2. **Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.**

```
# Stock A Prices

date <- c("2023-01-01", "2023-01-02", "2023-01-03")

stockA <- c(100, 105, 110)


# Calculate Daily Percentage Change

pct_change <- c(0, diff(stockA) / stockA[-length(stockA)] * 100)
```

```
# Display Percentage Change
print(pct_change)

# Create Bar Chart
barplot(pct_change,
        names.arg = date,
        col = "skyblue",
        main = "Daily Percentage Change in Stock A",
        xlab = "Date",
        ylab = "Percentage Change (%)")
```



3. **Build a table to display the stock price data for each company over the given period. Label the table elements.**

Stock Price Data

date <- c("2023-01-01", "2023-01-02", "2023-01-03")

stockA <- c(100, 105, 110)

stockB <- c(150, 152, 148)

stockC <- c(120, 118, 122)

Create Table

stock_data <- data.frame(

 Date = date,

 Stock_A = stockA,

 Stock_B = stockB,

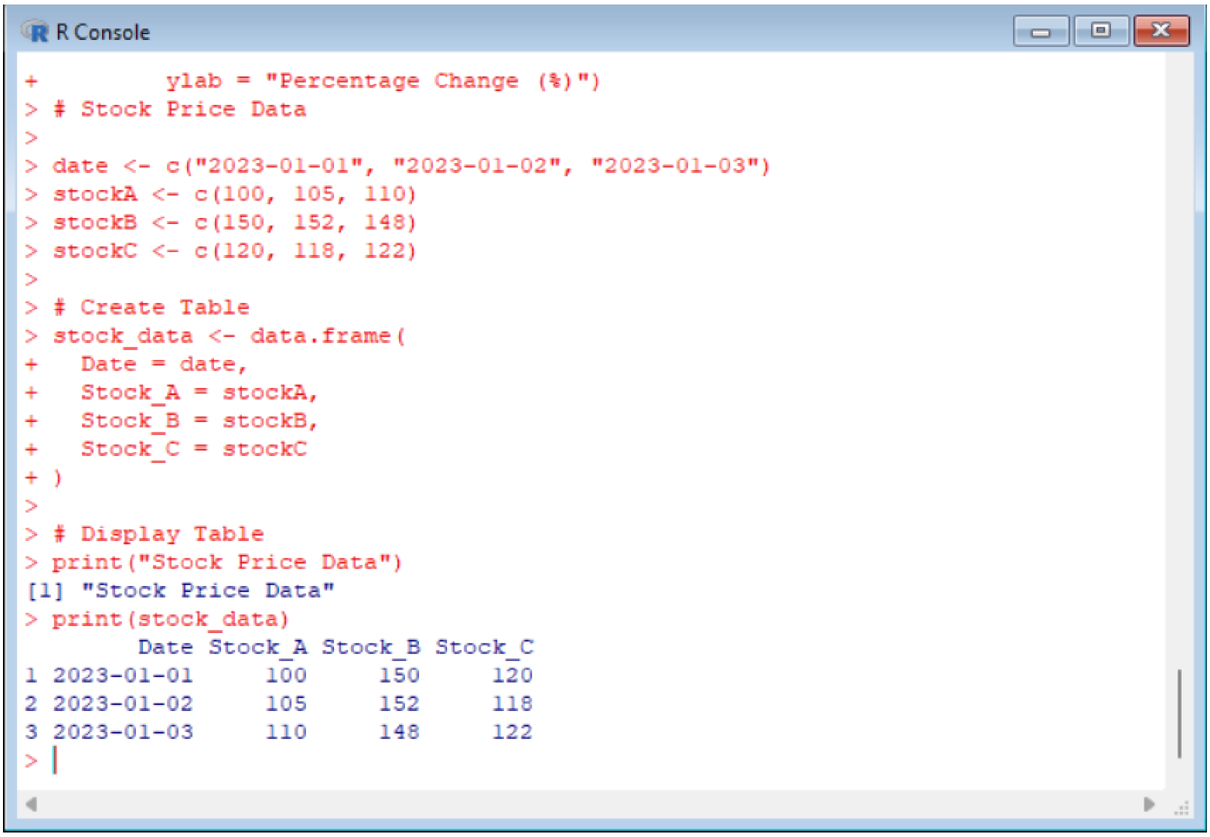
 Stock_C = stockC

)

Display Table

print("Stock Price Data")

print(stock_data)



4. **Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of stock prices.**

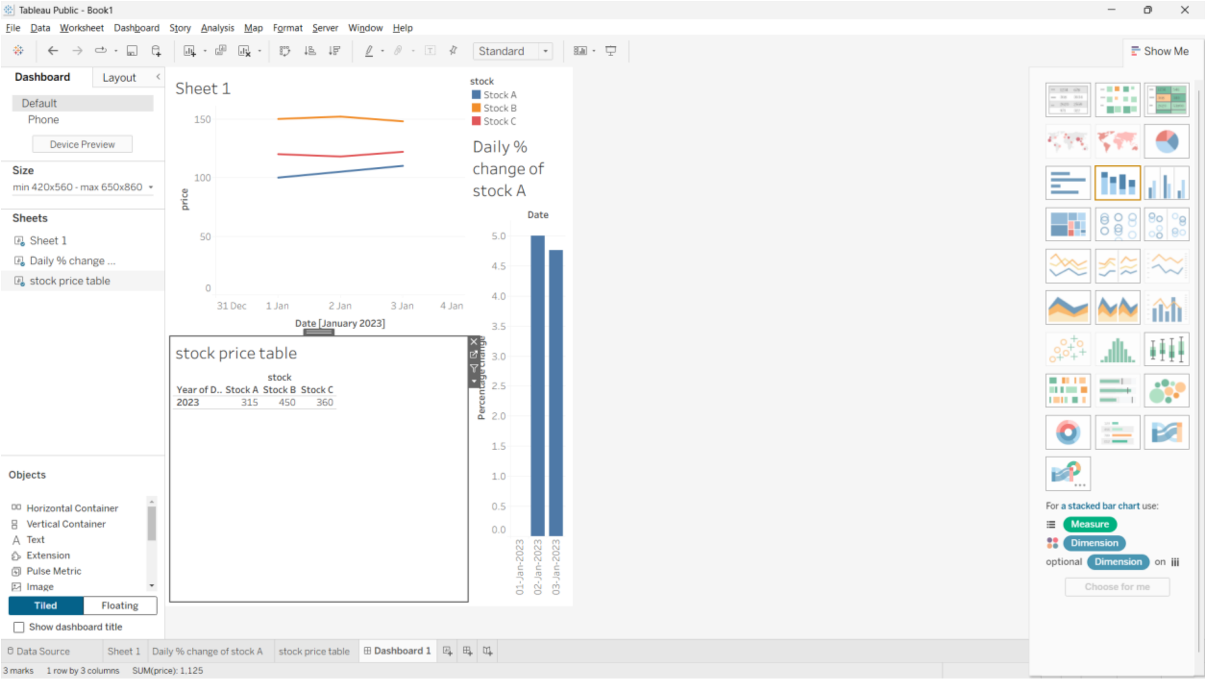


Tableau Public - Book1
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Standard
Show Me
Dashboard
Layout
Default
Phone
Device Preview
Size
min 420x560 - max 650x860
Sheets
Sheet 1
Daily % change ...
stock price table
Objects
Horizontal Container
Vertical Container
Text
Extension
Pulse Metric
Image
Tiled
Floating
Show dashboard title
Sheet 1
price
Date [January 2023]
stock
Stock A
Stock B
Stock C
Daily % change of stock A
Date
stock price table
Year of D.. Stock A Stock B Stock C
2023 315 450 360
01-Jan-2023
02-Jan-2023
03-Jan-2023
For a stacked bar chart use:
Measure
Dimension
optional Dimension on
Choose for me
Data Source
Sheet 1
Daily % change of stock A
stock price table
Dashboard 1
3 marks 1 row by 3 columns SUM(price): 1,125